PROJECT SYNOPSIS

Title: E-Jewels Full Stack Jewellery E-Commerce Website

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1. Project Overview

The Jewellery Shop Management System is a full stack web-based application developed to automate and manage the daily operations of a jewellery business. The system provides an integrated platform where the shop owner (Admin) can manage jewellery products, categories, inventory, customer details, orders, and billing efficiently. Customers can register, browse jewellery items, search products, add items to cart, place orders, and view their order history.

In the traditional jewellery business, most operations are carried out manually, which leads to inefficiency, data redundancy, and human errors. This project aims to digitize the entire process and provide a reliable, secure, and user-friendly system that enhances business productivity and customer satisfaction.

The application is developed using modern web technologies following a client–server architecture. It ensures secure authentication, proper authorization, and efficient data storage using a database management system.

2. Problem Statement

Most small and medium-scale jewellery shops still rely on manual systems to manage inventory, sales, billing, and customer records. These traditional methods involve maintaining registers and paperwork, which is time-consuming, error-prone, and difficult to manage as the business grows. Tracking stock availability, generating bills, managing customer information, and maintaining order history becomes complicated and inefficient.

Additionally, customers do not have an online platform to browse jewellery items or place orders remotely. This limits business growth and customer reach. Therefore, there is a strong need for a web-based Jewellery Shop Management System that automates business operations, reduces manual effort, minimizes errors, improves inventory control, and provides customers with an online shopping experience.

3. Scope of the Project

3.1 Objectives

- To design and develop a full stack web-based Jewellery Shop Management System.
- To automate product management, inventory control, sales, and billing.
- To provide a user-friendly interface for both admin and customers.

- To reduce manual work and improve data accuracy.
- To enhance customer experience through online browsing and ordering.
- To maintain secure access using authentication and authorization.

3.2 Features of the System

- Admin login and dashboard
- Jewellery product management (Gold, Silver, Diamond, etc.)
- Category-wise jewellery classification
- Customer registration and login
- Online jewellery browsing
- Search and filter functionality
- Add to cart and order placement
- Order tracking and history
- Billing and invoice generation
- Inventory and stock management
- Secure authentication and authorization

4. Existing System

In the existing system, jewellery shop operations are handled manually. Product details, inventory, sales records, and customer information are maintained using physical registers or spreadsheets. This system has several limitations:

- High chances of human errors
- Time-consuming data entry
- Difficulty in tracking inventory
- No centralized data storage
- No online facility for customers
- Poor security of data

5. Proposed System

The proposed system is a **web-based Jewellery Shop Management System** that replaces the manual process with a digital solution. All data is stored in a centralized database, and operations are handled through a secure web interface. The admin can easily manage products, inventory, customers, and orders, while customers can browse jewellery and place orders online.

Advantages of Proposed System

- Faster processing of operations
- Reduced manual effort
- Accurate data handling
- Better inventory management
- Improved customer satisfaction
- Secure and reliable system

6. Functional Requirements

Admin Functions

- Admin login and logout
- Add, update, delete jewellery products
- Manage jewellery categories
- Manage inventory stock
- View customer details
- View and manage orders
- Update order status
- Generate bills and reports

Customer Functions

- Customer registration
- Customer login and logout
- Browse jewellery items
- Search and filter products
- View jewellery details

- Add products to cart
- Place orders
- View order history

8. Hardware and Software Requirements

8.1 Hardware Requirements

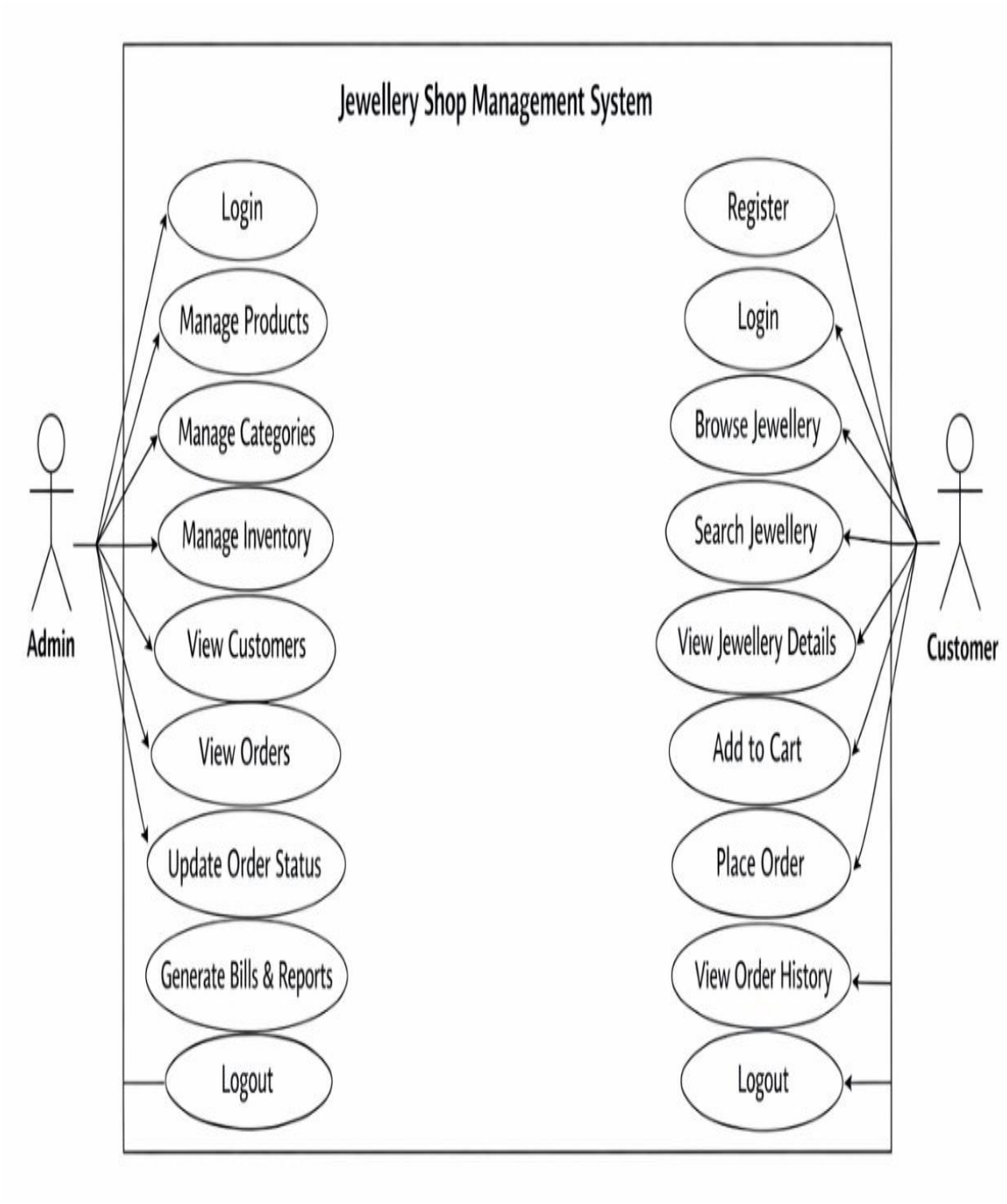
- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Hard Disk: 100 GB or more
- Internet connection
- Computer/Laptop

8.2 Software Requirements

- Operating System: Windows
- Frontend: HTML, CSS, JavaScript, React
- Backend: Node.js / Java
- Database: MongoDB
- IDE: VS Code
- Browser: Chrome

9. System Design

9.1 Business Use Case Diagram



10. Module Description

Admin Module

- Secure login
- Product and category management
- Inventory management and Order management
- Billing and reporting

Customer Module

- Registration and login
- Jewellery browsing
- Cart and order placement
- Order history tracking

11. Your Responsibilities

- Requirement analysis and planning
- System and database design
- Frontend UI development
- Backend API development
- Database integration
- Testing and debugging and Deployment
- Documentation and report preparation

12. Conclusion

The Jewellery Shop Management System provides an efficient, secure, and user-friendly platform for managing jewellery shop operations. By automating manual processes, the system improves accuracy, saves time, and enhances customer experience. This project demonstrates the practical application of full stack web development concepts and is suitable for real-world implementation.

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